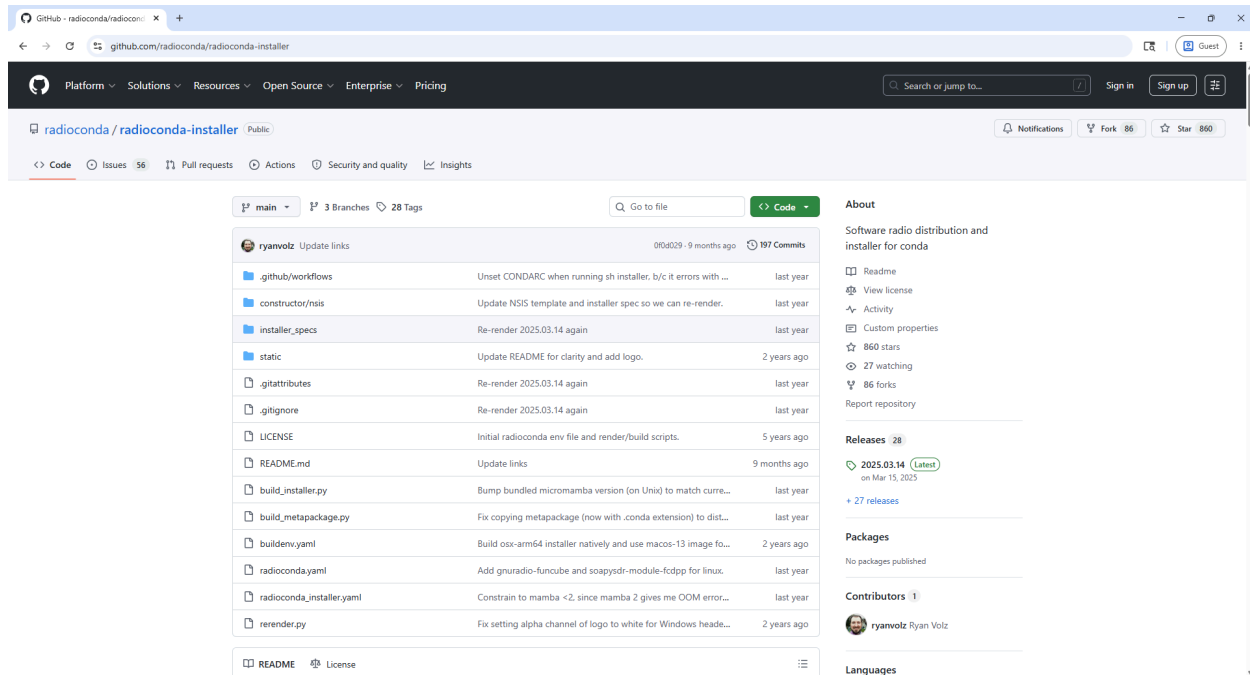


Getting up and running with Radioconda for SDRs- GUI Method for **Windows 10/11**:

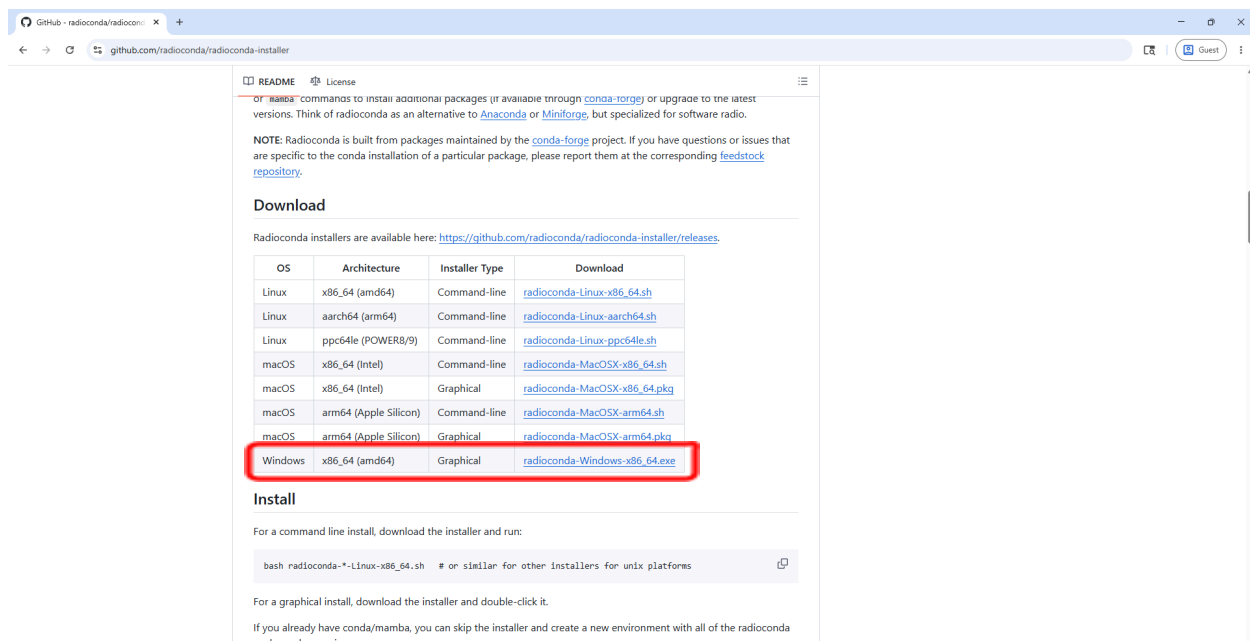
1: Search the GitHub page for Radioconda in your browser:

<https://github.com/radioconda/radioconda-installer>

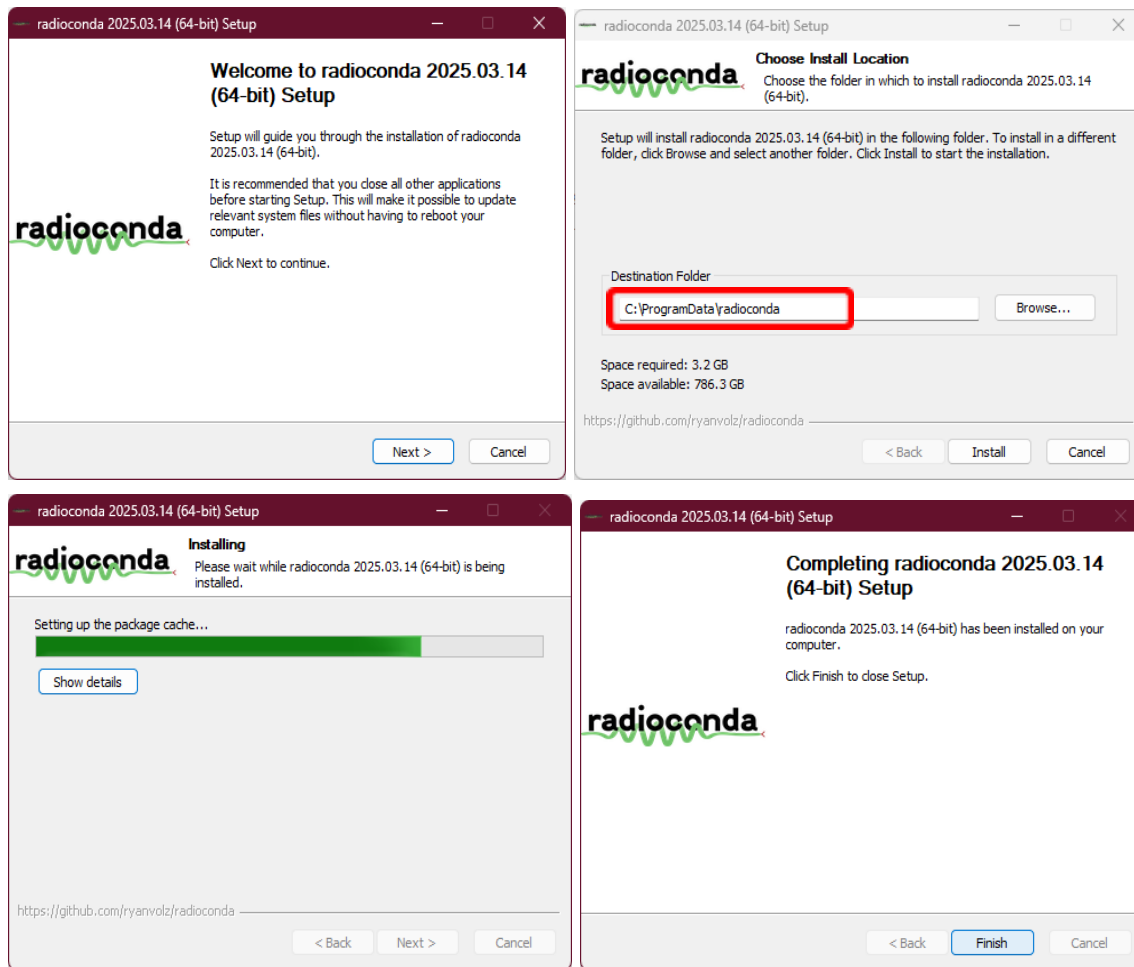


2: Scroll down to the download section and install the Windows exe installer file (~600 MB):

https://glare-sable.vercel.app/radioconda/radioconda-installer/radioconda-.*-Windows-x86_64.exe



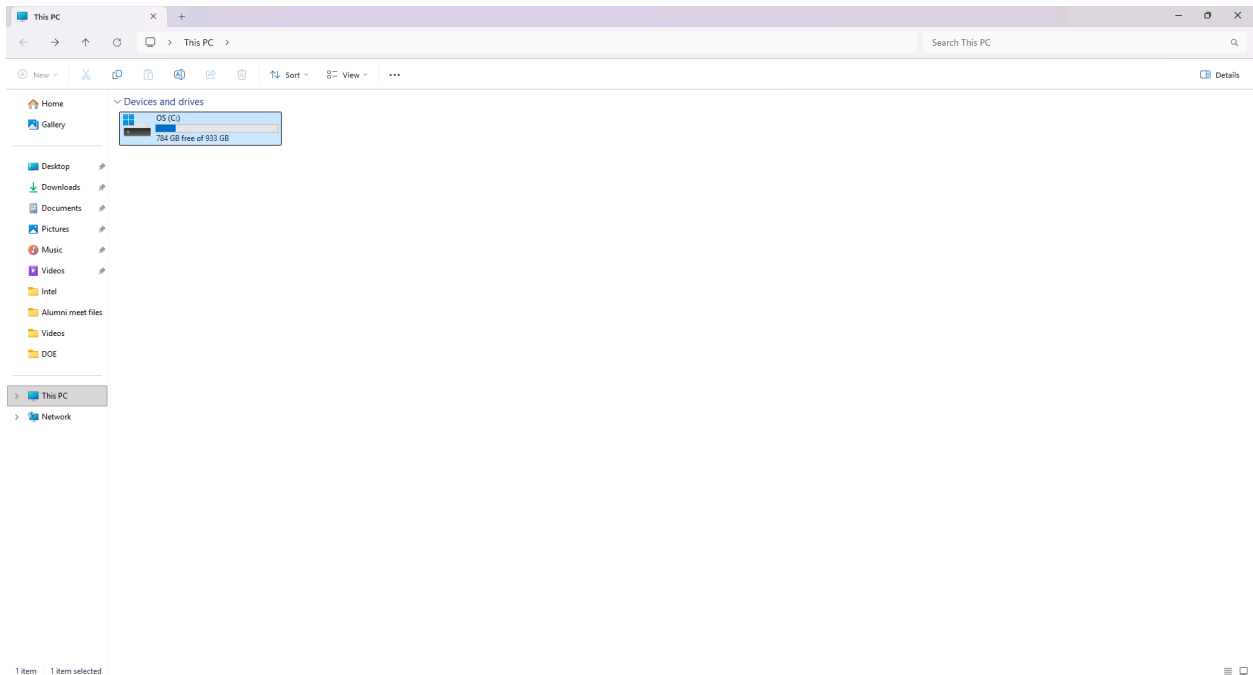
3: Run the downloaded exe installer file in the machine:
Follow the on-screen instructions and install for ALL USERS.



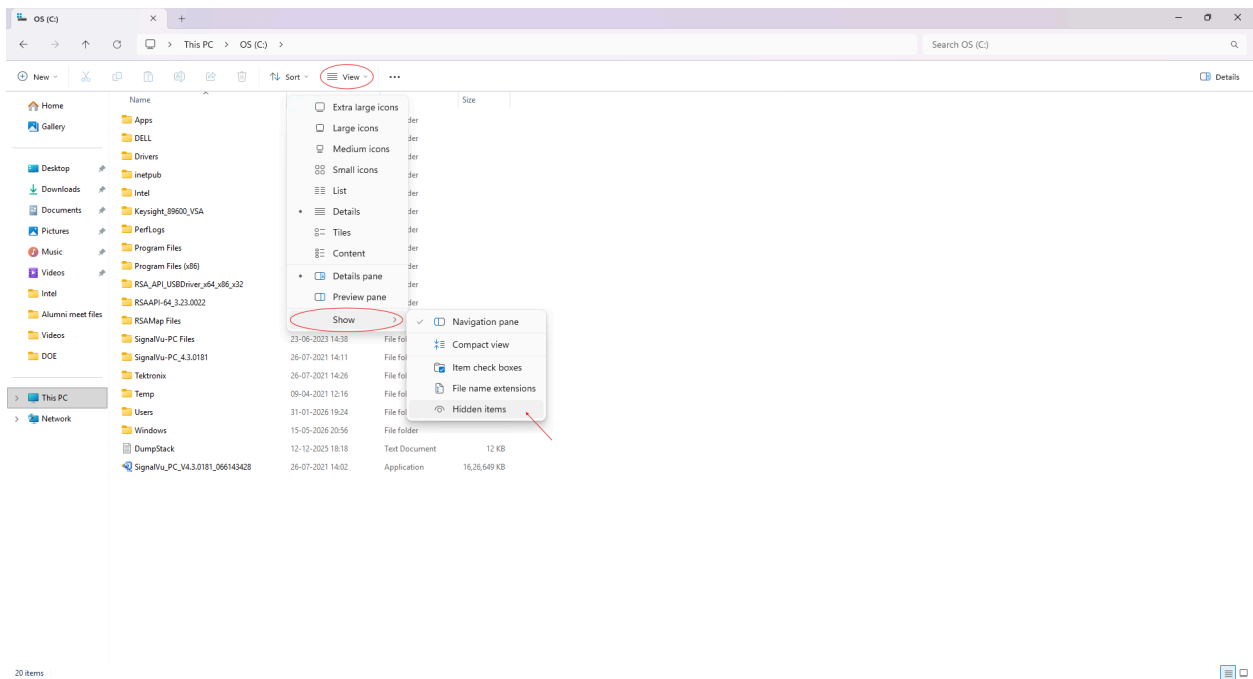
4: Setting up Radioconda to be run in PC for the SDRs

4a: Finding the file path and adding it to the PC Environment

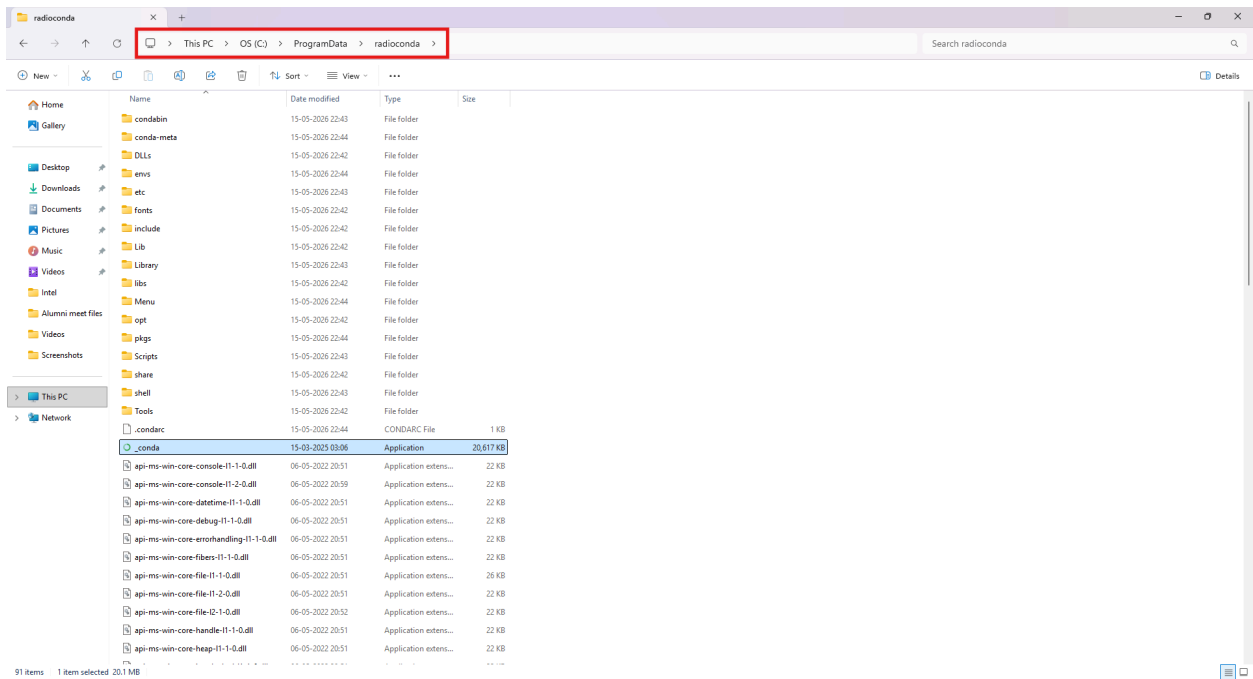
Open This PC -> <Your C: Drive>



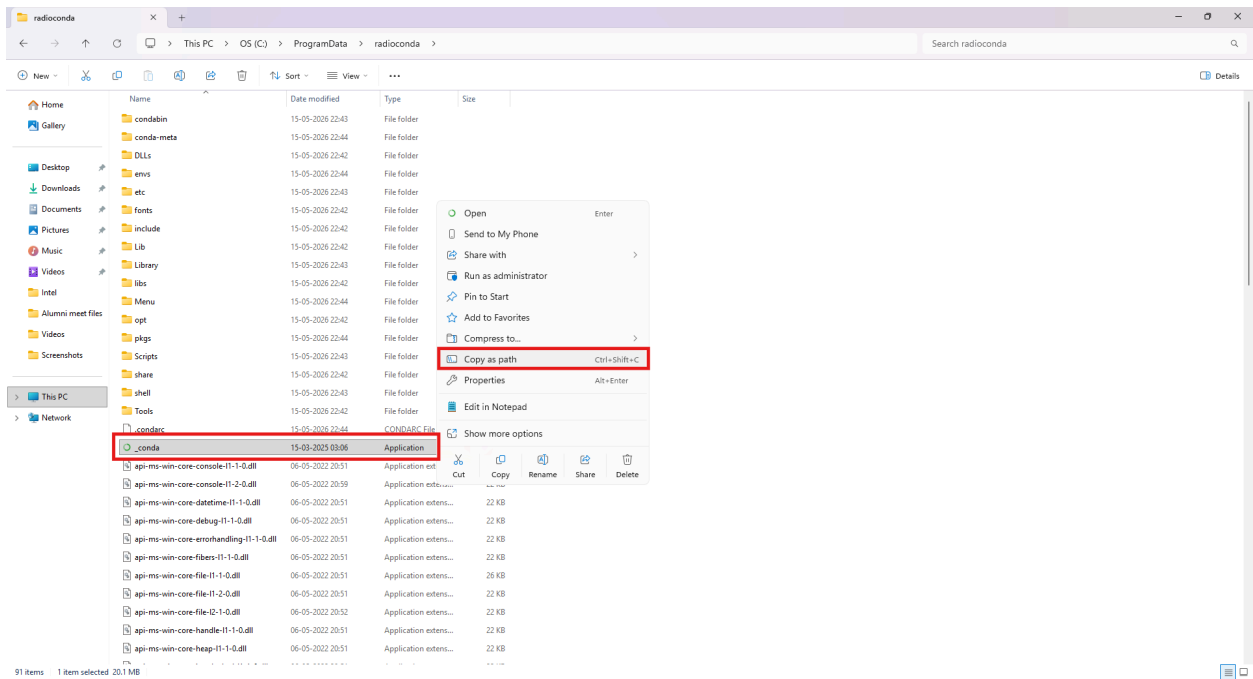
Enable “Show -> Hidden Items” under **View** menu



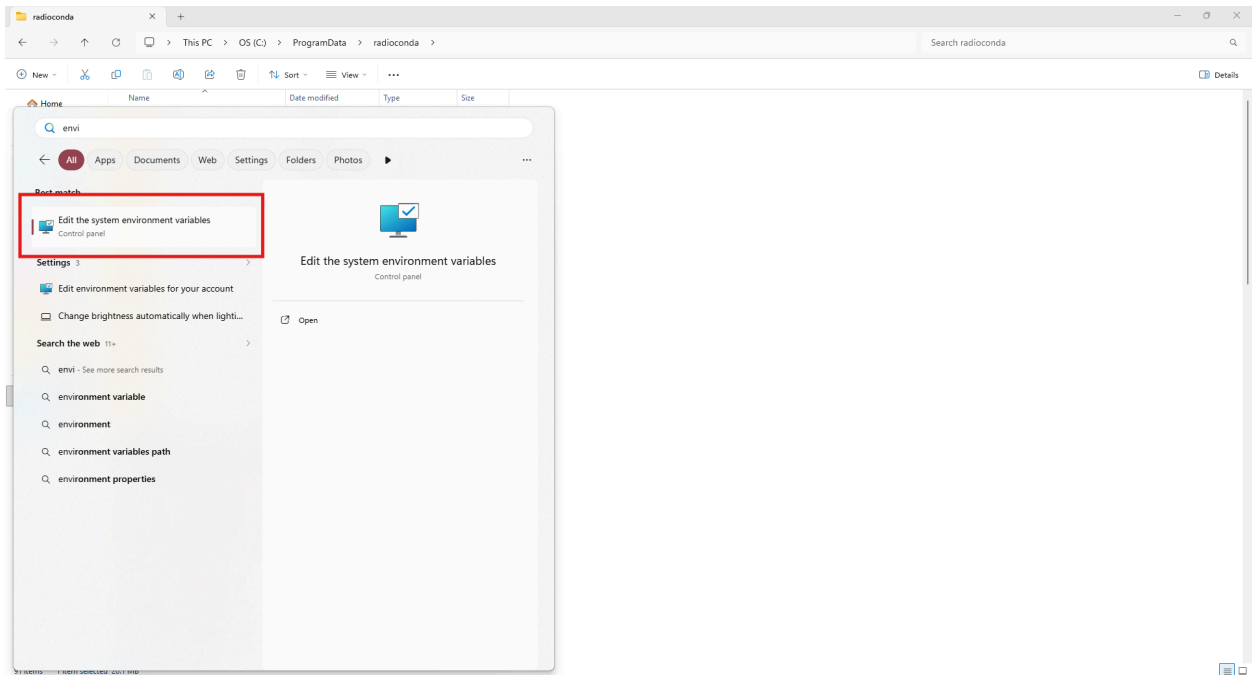
Now, Open **ProgramData > radioconda**



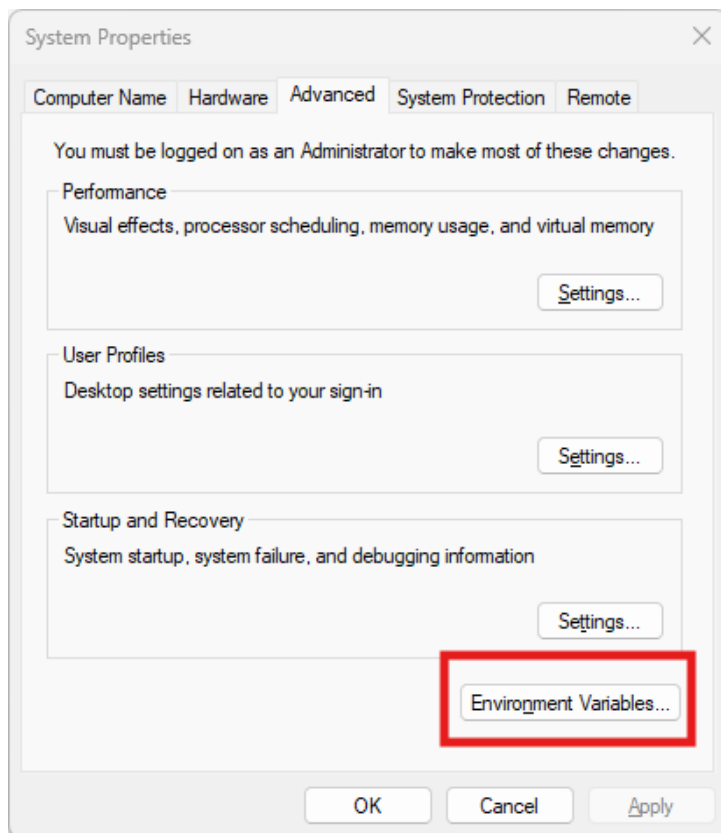
Right-click on **_conda** application file (~20 MB) and select **“Copy as path”**



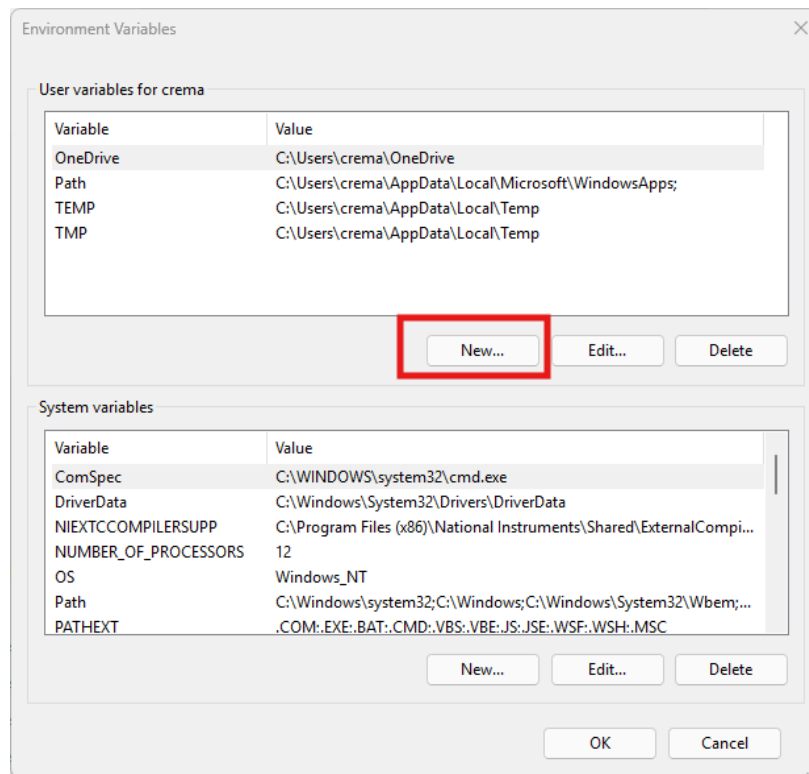
After copying the path, search Windows (**Winkey+S**) for **“Edit the system environment variables”**



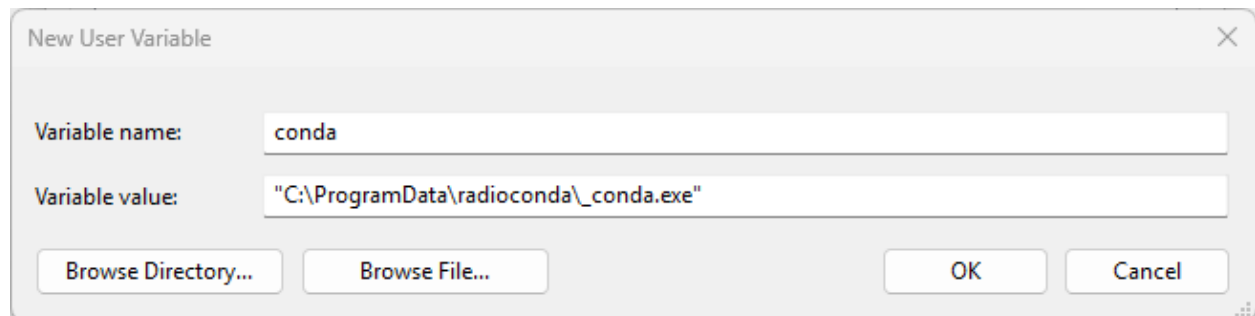
Select **“Environmental Variables”** in the System Properties dialog box



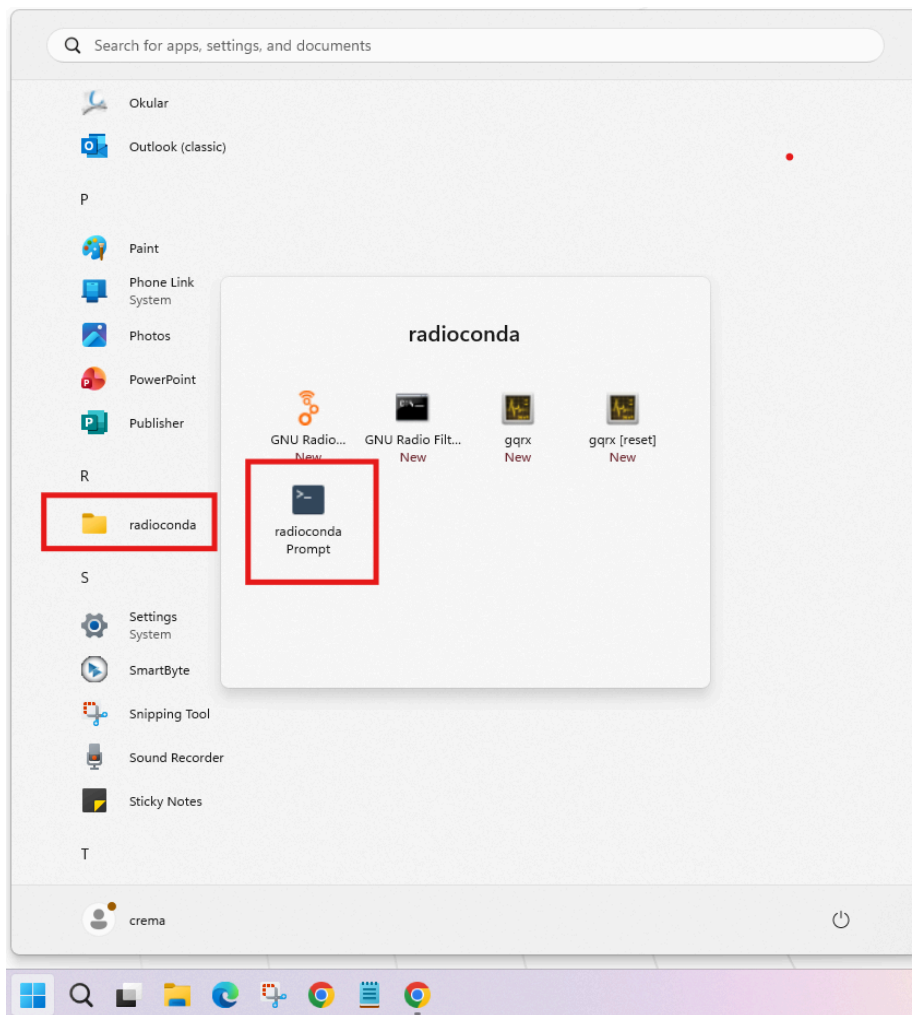
Under **Environment Variables** dialog box, press the **New** button within the **User Variables** section:



In the **New User Variable** dialog box, give a custom name (here: conda) and paste the copied path in the **Variable Value** and press **OK** and **Apply** in System Properties.



Go to **Start > All Programs/Apps > radioconda (folder)**

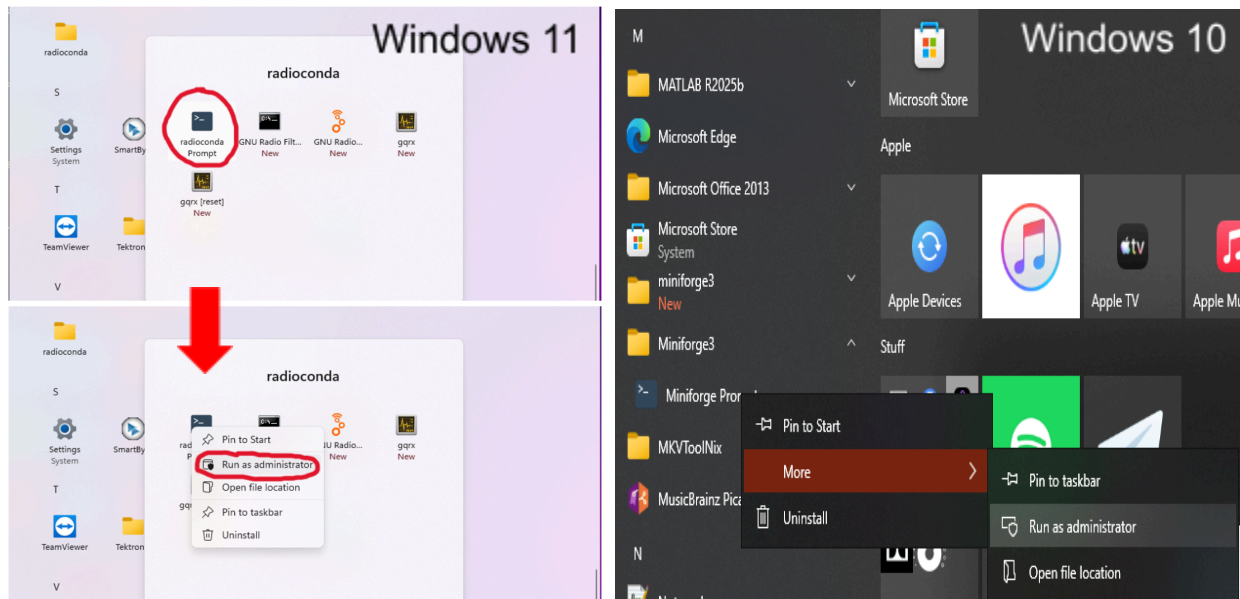


4b: Installing SDR Libraries in Radioconda:

For Ettus USRP B210 and NI USRP 2900 SDRs (UHD USRP B200 Family):

Go to **Start > All Programs/Apps(Just Scroll Down in Win 11) > radioconda (folder)**

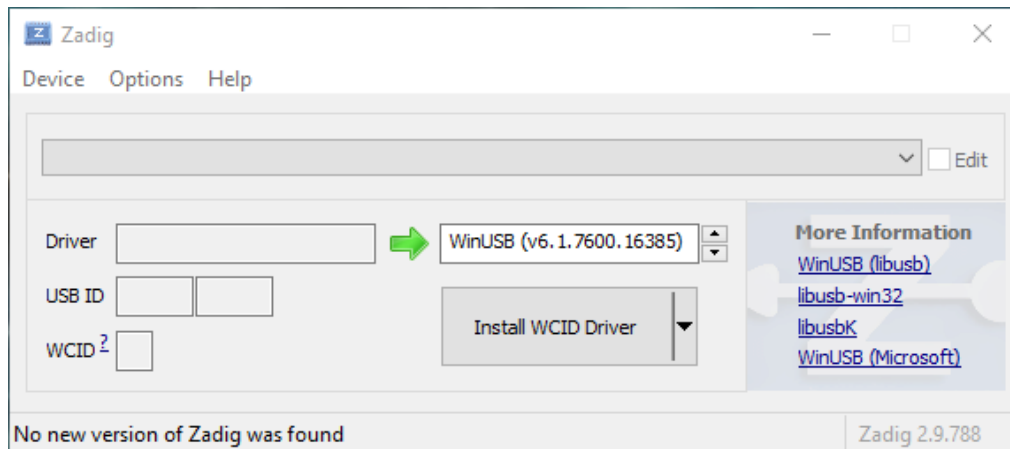
Right Click on **Radioconda Prompt** and select **Run as administrator (For Win 11) [or] More > Run as administrator (For Win 10)** (If prompted for admin permission, give **Yes**)



After the **command window** opens, type/paste the following command for the USRP Libraries:
uhd_images_downloader

```
Administrator: radioconda Prompt
(base) C:\Windows\System32>uhd_images_downloader
[INFO] Using Base URL: https://files.ettus.com/binaries/cache/
[INFO] Images destination: C:\ProgramData\radioconda\Library\share\uhd\images
[INFO] No inventory file found at C:\ProgramData\radioconda\Library\share\uhd\images\inventory.json. Creating an empty one.
31024 KB / 12924 KB (100%) x4xx_x440_fpga_default-g8e780be.zip
51611 KB / 51611 KB (100%) x4xx_x440_fpga_default-g8e780be.zip
19712 KB / 19712 KB (100%) x4xx_x440_fpga_default-g8e780be.zip
21598 KB / 21598 KB (100%) x4xx_x440_fpga_default-g8e780be.zip
81123 KB / 81123 KB (100%) e3xx_e310_sgl_fpga_default-g8e780be.zip
81118 KB / 81118 KB (100%) e3xx_e310_sgl_fpga_default-g8e780be.zip
10160 KB / 10160 KB (100%) e3xx_e320_fpga_default-g8e780be.zip
14218 KB / 14218 KB (100%) n3xx_n300_fpga_default-g8e780be.zip
18072 KB / 18072 KB (100%) n3xx_n310_fpga_default-g8e780be.zip
27214 KB / 27214 KB (100%) n3xx_n320_fpga_default-g8e780be.zip
88479 KB / 88479 KB (100%) b2xx_b2000_fpga_default-gc37b318.zip
88485 KB / 88485 KB (100%) b2xx_b200mini_fpga_default-gc37b318.zip
88870 KB / 88870 KB (100%) b2xx_b210_fpga_default-gc37b318.zip
88887 KB / 88887 KB (100%) b2xx_b200mini_fpga_default-gc37b318.zip
88887 KB / 88887 KB (100%) b2xx_common_fw_default-g7f78019.zip
88887 KB / 88887 KB (100%) usrp2_usrp2_fw_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp2_usrp2_fw_default-g8bea23d.zip
82415 KB / 82415 KB (100%) usrp2_n200_fpga_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp2_n200_fw_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp2_n210_fpga_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp2_n210_fw_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp1_usrp1_fpga_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp1_b100_fpga_default-g8bea23d.zip
88887 KB / 88887 KB (100%) usrp1_b100_fw_default-g8bea23d.zip
88887 KB / 88887 KB (100%) octoClock_octoClock_fw_default-g140000041.zip
84839 KB / 84839 KB (100%) usb_common_windrv_default-g140000041.zip
[INFO] Images download complete.
(base) C:\Windows\System32>
```


Also, install Zadig USB Driver Installer (<https://zadig.akeo.ie/>) from the below link:
<https://github.com/pbatard/libwidi/releases/download/v1.5.1/zadig-2.9.exe>



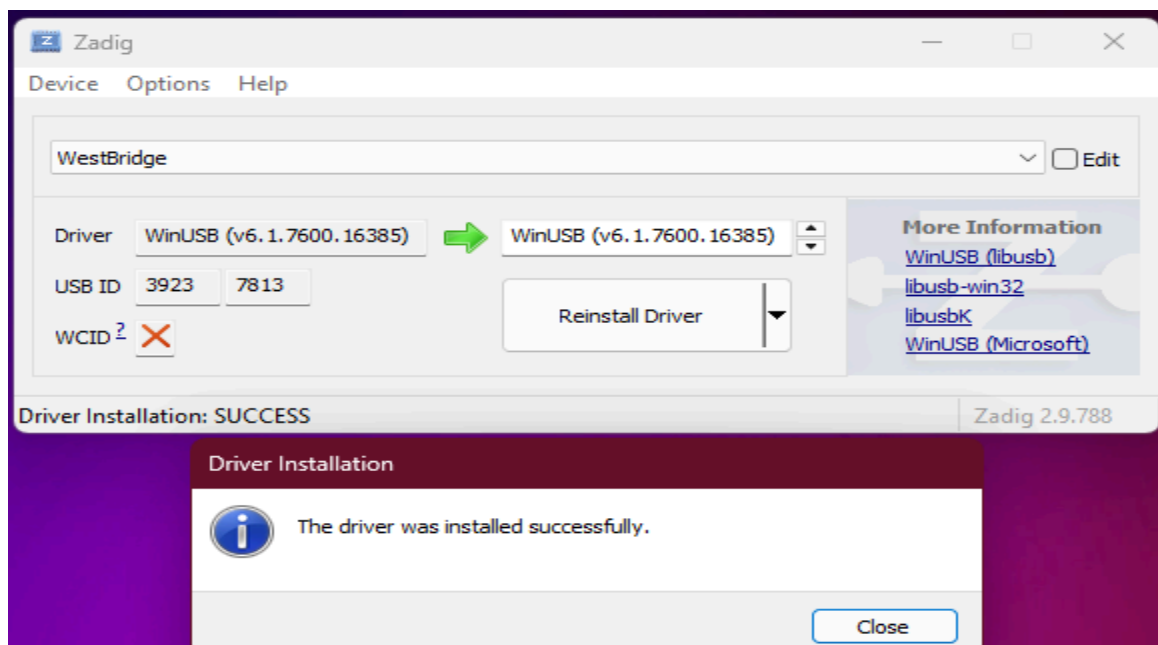
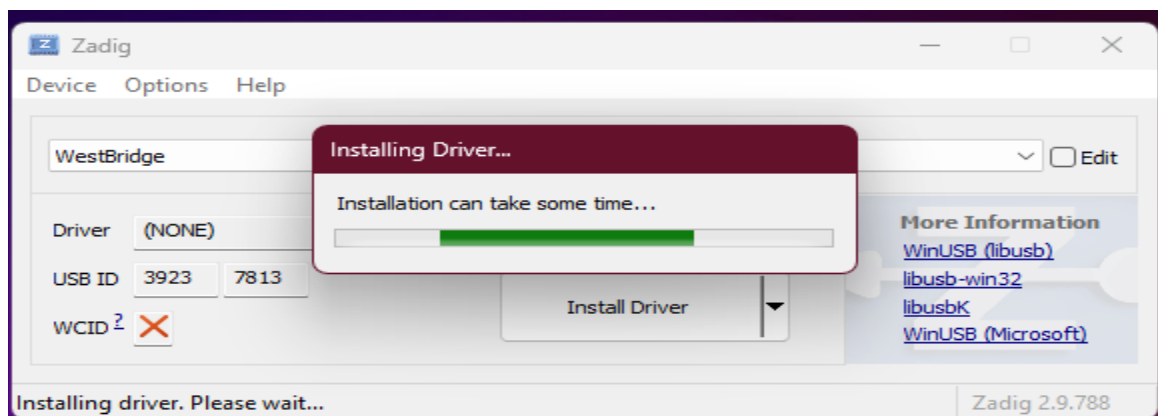
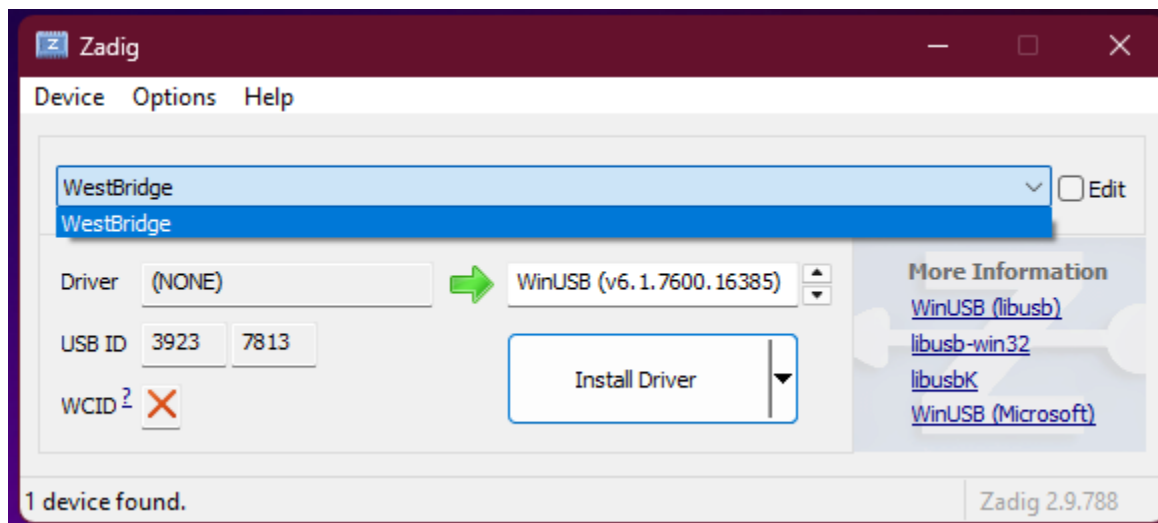
The above WinUSB Driver Installer is essential for running NI USRP 2900 and RTL-SDR Blog V4 SDRs in Windows 10/11 machines.

To check if the UHD USRP Devices are connected (**USB 3.0 Mandatory**), type the following command:

uhd_find_devices

```
Administrator: radioconda Prompt
(base) C:\Windows\System32>uhd_find_devices
[INFO] [UHD] Win32; Microsoft Visual C++ version 14.2; Boost_108600; UHD_4.8.0.0-release
No UHD Devices Found
(base) C:\Windows\System32>
```

If the device is not found (once you get your hands on the SDRs), open Zadig with the device connected and find the device “**WestBridge**” and install its driver (for NI 2900 (and B210, if needed)).



After installation, run the same command again:

```
Administrator: radioconda Prompt

(base) C:\Windows\System32>uhd_find_devices
[INFO] [UHD] Win32; Microsoft Visual C++ version 14.2; Boost_108600; UHD_4.8.0.0-release
No UHD Devices Found

(base) C:\Windows\System32>uhd_find_devices
[INFO] [UHD] Win32; Microsoft Visual C++ version 14.2; Boost_108600; UHD_4.8.0.0-release
[INFO] [B200] Loading firmware image: C:\ProgramData\radioconda\Library\share\uhd\images\usrp_b200_fw.hex...
[INFO] [B200] Loading firmware image: C:\ProgramData\radioconda\Library\share\uhd\images\usrp_b200_fw.hex...
-----
-- UHD Device 0
-----
Device Address:
  serial: 313469B
  name: 2ni2900
  product: B200
  type: b200
-----
-- UHD Device 1
-----
Device Address:
  serial: 313FA69
  name: MyB210
  product: B210
  type: b200
(base) C:\Windows\System32>
```

For further check-up and thorough functioning of the USRP SDRs, one can use the next command:

uhd_usrp_probe

```
Administrator: radioconda Prompt

(base) C:\Windows\System32>uhd_usrp_probe
[INFO] [UHD] Win32; Microsoft Visual C++ version 14.2; Boost_108600; UHD_4.8.0.0-release
[INFO] [B200] Detected device: B210
[INFO] [B200] Loading FPGA image: C:\ProgramData\radioconda\Library\share\uhd\images\usrp_b210_fpga.bin...
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Detecting internal GPSDO...
[INFO] [GPS] No GPSDO found
[INFO] [B200] Initialize CODEC control...
[INFO] [B200] Initialize Radio control...
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Setting master clock rate selection to 'automatic'.
[INFO] [B200] Asking for clock rate 16.000000 MHz...
[INFO] [B200] Actually got clock rate 16.000000 MHz.

Device: B-Series Device
-----
MBoard: B210
revision: 4
product: 2
name: MyB210
serial: 313FA69
FW Version: 8.0
FPGA Version: 16.0

Time sources: none, internal, external, gpsdo
Clock sources: internal, external, gpsdo
Sensors: ref_locked

-----
RX DSP: 0
Freq range: -8.000 to 8.000 MHz

-----
RX DSP: 1
Freq range: -8.000 to 8.000 MHz

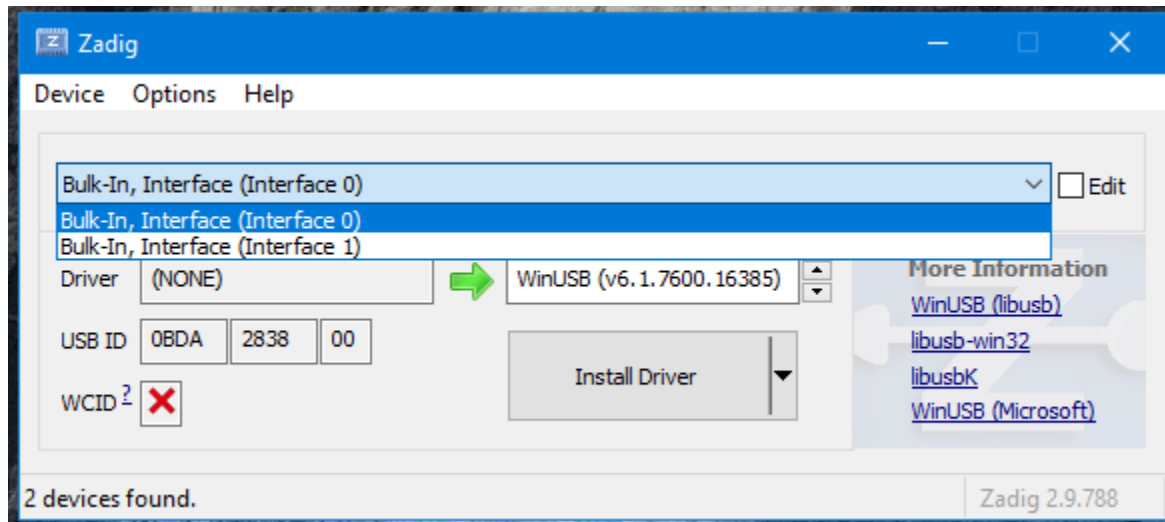
-----
RX Dboard: A
-----
RX Frontend: A
Name: FE-RX2
Antennas: TX/RX, RX2
Sensors: temp, rssi, io_locked
Freq range: 50.000 to 6000.000 MHz
Gain range PGA: 0.0 to 70.0 step 1.0 dB
Bandwidth range: 200000.0 to 56000000.0 step 0.0 Hz
Connection Type: IQ
Uses LO offset: No

-----
RX Frontend: B
Name: FE-RX1
Antennas: TX/RX, RX2
Sensors: temp, rssi, io_locked
```

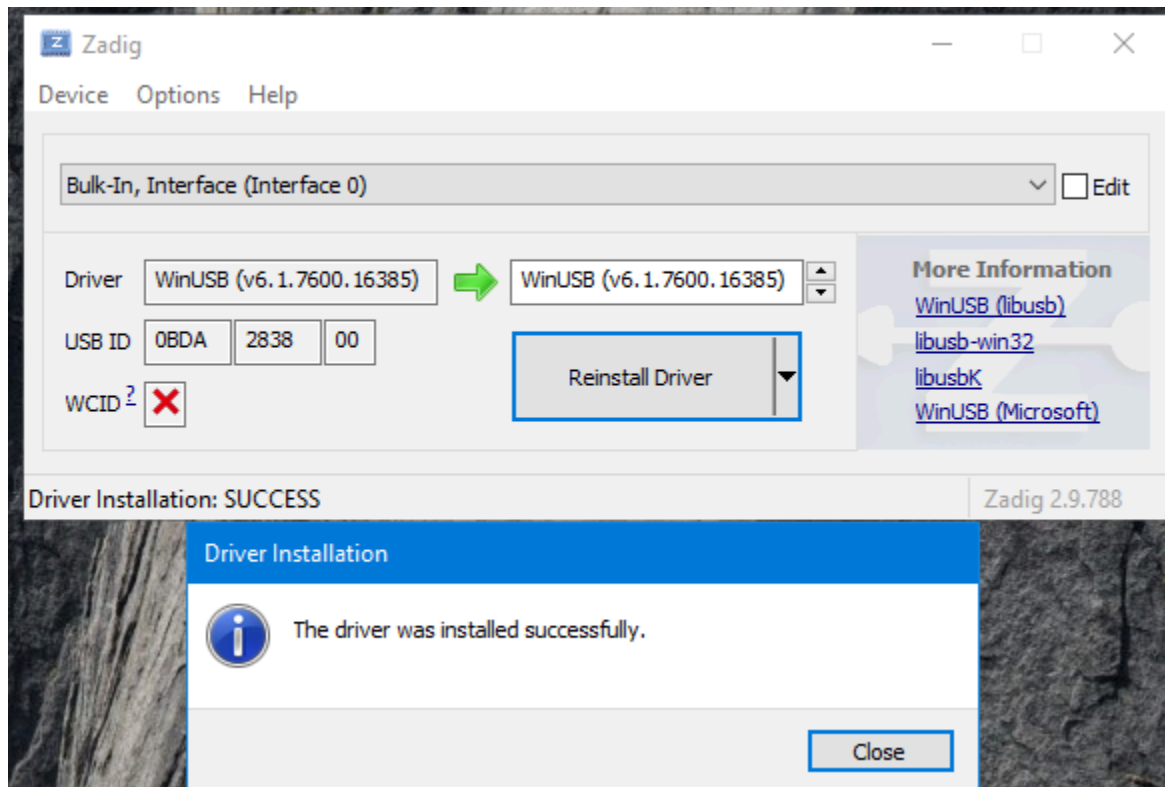
For RTL-SDR Blog V4 SDR:

After inserting the RTL-SDR Blog V4 in the USB port (USB 2.0 is fine), open Zadig (If prompted for admin permission, give **Yes**).

In Zadig's Dialog Box, select **Bulk-In, Interface (Interface 0)** and press **Install Driver**:



After successful installation, close Zadig:



Go back to the **Radioconda Prompt** command window. In the window, type the following command:

rtl_test

This command searches for samples to read while detecting the SDR. Press **Ctrl+C** to cancel the sample search immediately. You have successfully got your RTL-SDR up and running.

```
Administrator: radioconda Prompt
(base) C:\Windows\System32>rtl_test
Found 1 device(s):
  0: RTLSDRBlog, Blog V4, SN: 00000001

Using device 0: Generic RTL2832U OEM
Found Rafael Micro R828D tuner
RTL-SDR Blog V4 Detected
Supported gain values (29): 0.0 0.9 1.4 2.7 3.7 7.7 8.7 12.5 14.4 15.7 16.6 19.7 20.7 22.9 25.4 28.0 29.7 32.8 33.8 36.4 37.2 38.6 40.2 42.1 43.4 43.9 44.5 48.0 49.6
Sampling at 2048000 S/s.

Info: This tool will continuously read from the device, and report if
samples get lost. If you observe no further output, everything is fine.

Reading samples in async mode...
Signal caught, exiting!

User cancel, exiting...
Samples per million lost (minimum): 0

(base) C:\Windows\System32>
```